
Group08-Git Push The Beats

**Melody Dive
Vision Document**

Version <1.0>

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MD-VD-001-V1.0	

Revision History

Date	Version	Description	Author
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Vision (Small Project)

1. Introduction

1.1 Purpose

The purpose of this document is to collect, analyze, and define high-level needs and features of Melody Dive. It focuses on the capabilities needed by the stakeholders and the target users, and why these needs exist. The details of how the Melody Dive fulfills these needs are detailed in the use-case and supplementary specifications.

1.2 Scope

This Vision Document applies to the Melody Dive music streaming system, which will be developed by our team. The development team will build this client-server web application to provide users with an intuitive interface for searching, listening to, and managing songs.

The Melody Dive System will allow users to stream music online, like and save their favorite tracks, and share songs with others. The system will also enable users to view song lyrics, explore trending playlists.

1.3 References

None

2. Positioning

2.1 Problem Statement

The problem of	Many music streaming platforms nowadays, such as Zing MP3 and Spotify, contain a large number of advertisements.
affects	Music listeners who use these platforms.
the impact of which is	These frequent advertisements cause annoyance and time waste, interrupting users' listening experience.
a successful solution would	Provide a seamless and enjoyable listening experience, allowing users to enjoy music without advertisements.

2.2 Product Position Statement

For	Music enthusiasts and frequent listeners who want to enjoy songs conveniently on both desktop and mobile devices.
Who	Are seeking a free and user-friendly platform that provides an enjoyable listening experience without being interrupted by frequent advertisements.
The (product name)	is a website
That	Allow users to search, play, and organize their favorite songs with ease; it also provides full lyrics, trending charts, and AI-powered semantic search for intuitive discovery.
Unlike	Zing MP3 and Spotify
Our product	Offer an ad-free listening experience for regular users, and smart features like semantic search, delivering a seamless and enjoyable music experience.

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3. Stakeholder and User Descriptions

3.1 Stakeholder Summary

Name	Description	Responsibilities
Development team	Internal developers	Develop and monitor the product.
Ho Tuan Thanh	Supervisor	Provide feedback and guidance throughout the process.
Le Minh Tri	Music lover	Represents the needs and interests in listening to music.

3.2 User Summary

Name	Description	Stakeholder
Normal user	A regular user who can access and use basic features.	Self-represented
Premium user	A paid user who can use basic features and additional premium features.	Self-represented
Admin	The system administrator who monitors the website.	Self-represented

3.3 User Environment

Users are individuals who enjoy listening to music, and want to completely immerse themselves without being interrupted by advertisements.

Users can access the website anytime in the day, using either desktop or mobile devices.

Users should have a bank account, so that they can pay for premium accounts.

3.4 Summary of Key Stakeholder or User Needs

Need	Priority	Concerns	Current Solution	Proposed Solutions
Quick access to music	High	Users waste time searching manually	Users must search on Youtube, Google,... or buy CDs	Search by title, genre, composer or lyrics using semantic AI

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Instant playback	High	No immediate access to songs	Users must purchase physical CDs or download files manually, then use a media player to play songs. This process takes time and limits flexibility	One-click play with smooth streaming and playback controls
Lyrics availability	High	It takes more time to find accurate lyrics	Users search lyrics on third party-sites	Display synced lyrics directly with the song, allowing users to follow along in real time and better understand the meaning of the song
Access to exclusive songs	High	Users must pay per song or watch ads on other platforms to access premium content, which is time-consuming and fragmented	Users purchase songs individually or rely on ad-supported platforms	Premium users can access all songs on the website with a one-time payment, ad-free and unlimited.
Organize favorite songs	High	No centralized way to save songs	Users save links or files manually	Create playlists and collections in user account
View song details	Medium	Lack of context about song origin	Users search manually for composer or release info	Show full metadata: composer, release date, genre,...
Unlimited playlist creation	Medium	Regular users hit playlist limits	Users manually split playlists, delete old ones, or save songs elsewhere (e.g., notes, browser bookmarks) to manage their collections	Premium users can create unlimited playlists and organize music freely within the system
Comment and interact	Medium	No space to express opinions	Users post on external forums or social media	Allow comments on songs with moderation tools
Report handling	High	Harmful content exists	Admins manually review reports	Dashboard to manage reports, delete content, and ban users

3.5 Alternatives and Competition

1. Spotify

- Strengths:
 - Easy-to-use interface, strong personalized recommendations, wide global catalog.

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- Huge user base, strong brand recognition, advanced data analytics.
- Weaknesses:
 - Ads disrupt free tier, limited audio quality, subscription costs.
 - Local/independent artists feel underrepresented.
- 2. **Apple Music**
- Strengths:
 - High-quality audio including lossless and spatial formats, seamless integration across Apple devices, curated editorial playlists.
 - Premium positioning within Apple's ecosystem, loyal customer base, ability to secure exclusive releases.
- Weaknesses:
 - No free tier, discovery features less engaging.
 - Limited reach outside Apple's ecosystem, weaker presence in Android-dominated markets.
- 3. **Homegrown solutions:** Building personal playlists using downloaded MP3s or local media players.
- Strengths: Full control over music ownership, no recurring fees.
- Weaknesses: Limited accessibility across devices, lack of social or discovery features, manual updates required.

Our music streaming website provides a free tier with fewer interruptions, designed with a clean, user-friendly interface and responsive layout that works well on both desktop and mobile. It focuses on music discovery through curated playlists and community features, while giving more visibility to local and independent artists.

4. Product Overview

This section provides a high level view of the product capabilities, interfaces to other applications, and system configurations.

4.1 Product Perspective

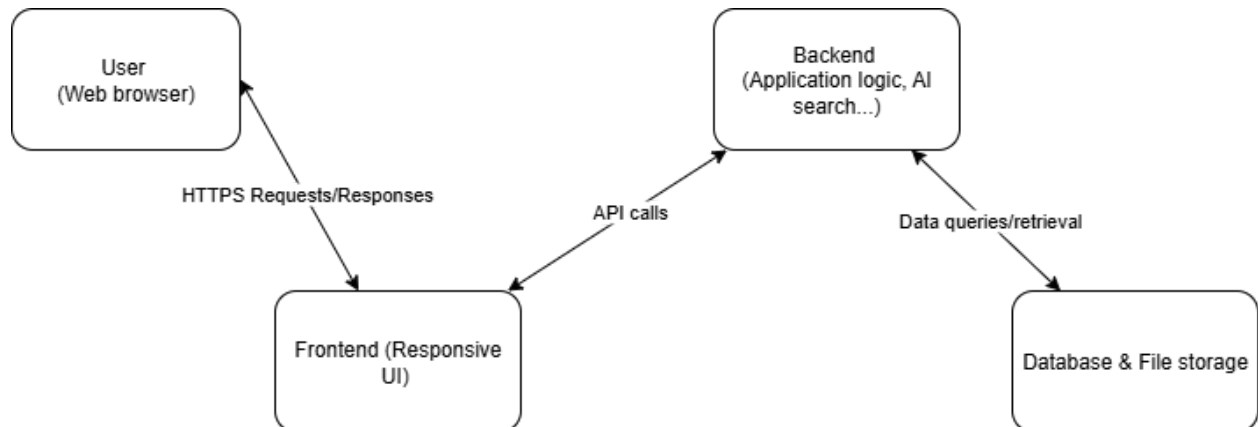
Melody Dive is a self-contained, web-based music streaming service. It is designed to be a standalone application, accessible to users through standard web browsers on both desktop and mobile devices, eliminating the need for any client-side software installation.

The system's architecture consists of three primary components:

- Frontend (Client): A responsive user interface that runs in the user's web browser, providing all the interactive elements for music playback, search, and playlist management.
- Backend (Server): The core of the application, responsible for managing user accounts, processing search queries (including the AI-powered semantic search), handling song metadata, and streaming audio files to the user.
- Data Storage: A combination of a database to store structured data (user profiles, song information, playlists, comments) and a file storage system for the actual music tracks.

The product operates within the user's internet environment and interacts with external systems only for standard web functionalities. There are no dependencies on other software products for its core features to function. A simplified block diagram of the system is as follows:

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4.2 Assumptions and Dependencies

- The system needs to run 24/7.
- The UI needs to be responsive. Users can access the system via a modern web browser (e.g., Chrome, Firefox, Edge) and through mobile devices.
- Integrating with local payment providers for premium registrations (e.g., VNPay).
- The customer base is expected to be small to medium scale in the first release.

5. Product Features

5.1 Normal Users

- Account management: Users create an account, log in, and reset the user's password.
- Play music: Users can play and rewind songs.
- Provide Lyrics: Lyrics of music are provided to users.
- Playback control: Users can listen to music in playback order and playback mode (repeat once, repeat continuously, play in order).
- Download: Users can download songs from the website.
- Searching: Users can search for songs and refine results with filters (genre, composer).
- Semantic Search (AI feature): Users can search for songs by phrases (for example, the line “le doi ban chan buoc di tren con pho dai” is a lyric in the song “Tuyet Yeu Thuong” by Young Uno).
- Show trending songs: Users can view the most recently listened songs.
- Search History: Save search history for quick access for users.
- Group of songs: View albums and song categories by genre, language..
- Favorites and Collections: Save users’s favorite songs and organize them into personal Playlists (limited to regular accounts).
- Comment: Users can post a comment about a song.
- Report: Users can report inappropriate or harmful comments/songs.
- Users can upgrade their account from “Normal User” to “Premium User”.

5.2 Premium Users

- Premium Users can use all features of regular users.
- Premium Users can listen to all songs, including premium ones.
- Premium Users can save songs to unlimited personal playlists.

5.3 Administrator

- Song management: Admins can add, edit, delete and categorize songs, albums, genres.
- Premium account management: Admins can modify the price of Premium accounts (for example: 1 month cost 49.000VNĐ, 2 monthly premium cost 79.000 VNĐ,..).
- Report management: Admins can review and process reports from users.
- Ban users and deletes: Admins can delete inappropriate comments/songs. Ban malicious users.

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6. Non-Functional Requirements

6.1 System Requirements

- The system shall encrypt sensitive data during storage and transmission using secure and widely accepted methods, backup files shall be retained for 30 days and stored in encrypted format.
- Data recovery shall be possible within 15 minutes after failure.
- The website implements advanced security features such as Captcha and multi-factor authentication via email, SMS, or phone calls.
- The system shall provide an intuitive and user-friendly interface with consistent navigation, clear labeling, and minimal user training required.

6.2 Performance Requirements

- The system works smoothly across popular web browsers (Chrome, Firefox, Safari) and on both desktop and mobile devices.
- Pages should load within 2 seconds on local hardware.
- Song search (including semantic search) returns results within 1 second for datasets up to 500 songs.
- User history and playlists shall persist across sessions and the system can store at least 100 MB of data per normal user.
- The website supports at least from 5 to 10 simultaneous users for demo purposes without system crash.

6.3 Environmental Requirements

- The system shall be adaptable to run on a small server in case it is deployed online in the future.